

Predictive production system and supply chain automation to reduce waste at Panadería Victoria

Predictive production system and supply chain automation to reduce waste at Victoria Bakery Ayrton Eduardo Wilfredo Quiñones Jara 1 ,
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Abstract Food waste and inefficient supply chain management in artisanal and semi-industrial bakeries generate significant economic losses. The objective of this study was to implement a predictive production and automation system based on Artificial Intelligence and Business Intelligence tools to reduce production waste at Panadería Victoria (Pacasmayo, Peru) by at least 20%. The research presented a quantitative approach and a pre-experimental design, supported by the development of a decoupled architecture using PostgreSQL, FastAPI, React, and n8n. Machine Learning techniques were applied, evaluating seven predictive algorithms (Random Forest, Linear Regression, Gradient Boosting, SARIMA, Prophet, Neural Networks, and an Ensemble model) trained with 25 independent calendar and weather variables. The results

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Keywords: automation, supply chain, machine learning, production losses, demand forecasting

Abstract Food waste and inefficient supply chain management in artisanal and semi-industrial bakeries generate significant economic losses. The objective of this study was to implement a predictive production and automation system based on Artificial Intelligence and Business Intelligence tools to reduce production waste at Victoria Bakery (Pacasmayo, Peru) by at least 20%. The research featured a quantitative approach and pre-experimental design, supported by the development of a decoupled architecture using PostgreSQL, FastAPI, React, and n8n. Machine Learning techniques were applied evaluating seven predictive algorithms (Random Forest, Linear Regression, Gradient Boosting, SARIMA, Prophet, Neural Networks, and an Ensemble model) trained with 25 independent calendar and weather variables. Expected results indicated that the system successfully automated daily production suggestions and the issuance of purchase orders upon detecting inventories below safety limits. It is concluded that the implementation of predictive models optimizes inventory planning and supports data-driven decision-making. **Keywords:** automation, demand forecasting, machine learning, production waste, supply chain

1 **Introduction** Food waste throughout the supply chain constitutes a first-order challenge both environmentally and economically. Globally, the management

1 **Introduction** Food waste along the supply chain constitutes a first-order challenge both environmentally and economically. Globally, inefficient management of perishable products, particularly in the bakery industry, leads to high loss rates due to demand volatility and the short shelf life of products (Barka-Papadimitriou et al., 2025; Kroupová, 2024). In artisanal and semi-industrial bakeries, such as Panadería Victoria in Pacasmayo (Peru), production projections have historically relied on heuristic methods based on staff experience. This empirical practice typically incorporates arbitrary safety margins ranging between 10% and 25% above average sales, systematically generating overproduction or, in adverse scenarios, underproduction and stockouts that affect the service level.

Digitalization and data analytics have emerged as transformative tools in the food industry. The adoption of Machine Learning techniques and Artificial Intelligence allows for the extraction of non-linear patterns hidden in historical sales data and exogenous variables (Feng et al., 2025; İçöz, 2024; Wafi, 2025). Recent research maintains that demand forecasting through advanced algorithms outperforms

exogenous variables (Feng et al., 2025; İçöz, 2024; Wafi, 2025). Recent research maintains that demand forecasting using advanced algorithms significantly outperforms traditional statistical methods (such as simple moving averages), providing resilience and optimization to inventory management and logistics operations (Feizabadi, 2022; Samuels, 2025). In the specific context of the retail and food marketing sector, several studies

support the effectiveness of integrating contextual variables (such as holidays and weather factors) into predictive models. Authors such as Huber and Stuckenschmidt (2020) and Wikamulia et al. (2023) demonstrated that decision tree-based models, such as Random Forest or XGBoost, substantially increase the accuracy of demand forecasting when considering calendar events. Likewise, Concha (2026) and Windrasari et al. (2023) applied Machine Learning for process standardization and predictive sales analysis in gastronomic establishments, corroborating that these tools reduce operational uncertainty.

Beyond isolated modeling, current literature emphasizes the need for comprehensive systems. The connection between demand forecasting, dashboards, and supply chain automation forms a reactive and proactive ecosystem that prevents waste starting from raw material procurement (Henaö, 2023; Mishra, 2024; Monferdini, 2024; Pandey, 2023). Despite these advances, there is a notable scientific gap regarding the implementation of these integrated architectures in Small and Medium-sized Enterprises (SMEs) in the bakery sector within regional contexts of Latin America, where technological budget is often a restrictive factor.

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The technological budget is usually a restrictive factor. In response to this problem, the purpose of this research is to implement a predictive production and supply chain automation system based on Machine Learning algorithms and Business Intelligence tools. The main objective is to reduce production waste at Panadería Victoria by a percentage of no less than 20% compared to its historical level, guaranteeing a statistical confidence level of 95% and improving replenishment through automated processes.

2 Materials and methods

The study was framed under a quantitative approach, of an applied level and pre-experimental design, focusing on the analysis of transactional time series. The study setting was Panadería Victoria, located in Pacasmayo, La Libertad Region, Peru. The study population comprised the comprehensive set of historical records of daily sales, production waste, recipes (technical sheets), and inventory levels of critical supplies. The analyzed sample covered a continuous historical period that included simulated and real daily operational data to evaluate predictive efficacy.

2.1 System Architecture

To ensure scalability and availability, the system was structured using a local microservices architecture orchestrated with Docker Compose. It employed

To ensure scalability and availability, the system was structured using a local microservices architecture orchestrated with Docker Compose. PostgreSQL was employed as the relational database management system (visualized using pgAdmin 4). The physical schema was conceptually divided into dimensional tables (catalogs

of products, suppliers, weather) and fact tables (sales, waste, purchase orders). The analytical core and the RESTful API were developed in Python (FastAPI), implementing SQLAlchemy as an ORM. The operational user interface was built as a Single Page Application (SPA) in React and Vite, designed to support the Point of Sale (POS) and on-site production logging.

2.2 Feature Engineering and Study Variables Data processing involved an exhaustive Feature Engineering phase. From the timestamp (date) of each transaction, 25 independent predictor variables (features) were extracted and generated to train the models. These variables were grouped into:

- Temporal: Day of the week, month, day of the month, day of the year, and boolean flags for weekends.
- Events and Holidays: Binary indicators for Peruvian national holidays, holiday eves, post-holiday days, and specific commercial events (Christmas, Fiestas Patrias, Mother's Day).
- Meteorological: HTTP API integration with Open-Meteo for Pacasmayo, obtaining daily average temperature, presence of rain, and encoded weather conditions.
- Historical (Lags and Rolling Windows): Lagged demand from 1 and 7 days prior, as well as moving averages of sales from the last 7 and 30 days, allowing for the capture of inertial trends.

2.3 Machine Learning Modeling

such as 7-day and 30-day moving sales averages, allowing for the capture of inertial trends.

2.3 Machine Learning Modeling The artificial intelligence engine was not restricted to a single algorithm but rather implemented a competitive dynamic pipeline. For each product in the catalog, the training module processed and compared seven modeling architectures described in Table 1. The system automated the selection of the "best model" based on the minimization of the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE), ensuring the highest possible precision adapted to the specific behavior of each item.

No. Evaluated Algorithm Implementation Library Main Operational Characteristic

1 Random Forest scikit-learn Ensemble of decision trees to capture

No.	Evaluated Algorithm	Implementation Library	Main Operational Characteristic
			nonlinearity.
2	Linear Regression	scikit-learn	Baseline for linear trend detection.
3	Gradient Boosting	scikit-learn	Iterative sequential optimization of residual error.
4	SARIMA	statsmodels	Pure statistical seasonality treatment.
5	Prophet	prophet	Additive decomposition specialized in irregular holidays.
6	Neural Network (MLP)	scikit-learn	Multilayer perceptron for highly abstract patterns.
7	Hybrid Ensemble	scikit-learn	Voting/averaging combination to mitigate overfitting.

2.4 Supply Automation and Logical Control The supply chain was integrated with the n8n (Node-Based Workflow Automation) platform. The methodological protocol was defined by executing a daily temporal trigger at 8:00 AM. The engine queried the FastAPI API (alerts endpoint) to identify the list of critical inputs whose physical inventory, projected through sales forecasting, fell below the safety stock level. In such a scenario, the

The system applied a business rule to locate the supplier with the lowest unit rate and automatically generated a suggested purchase order with a "pending" status, using the double stock replenishment formula. Finally, a document alert in PDF was dispatched to the administrative team.

3 Results and discussion (This section will be formally completed once the experimental phase of the ongoing research is concluded, contrasting the Machine Learning model projections against the empirical volumetric records obtained in the bakery field tests).

Conclusions The structural and methodological development presented confirms the viability of modernizing the production environments of artisanal bakeries through advanced, low-licensing-cost technologies. The structuring of the multi-algorithm predictive pipeline, combined with the data orchestration architecture using n8n and FastAPI, establishes a solid technological framework capable of reacting in anticipation of demand volatility

influenced by exogenous factors. The correct implementation of this ecosystem ensures strict control over supply decisions, charting the operational path to significantly reduce food waste and improve the corporate sustainability margin at Panadería Victoria.

Acknowledgements

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Acknowledgments Gratitude is extended to the authorities and faculty members of the Escuela de Ingeniería de Sistemas at the Universidad Nacional de Trujillo for their academic guidance in the development of this research work.

Conflict of interest The author declares that there is no conflict of interest related to the subject matter of the research work presented.

Source of financing

The author did not receive any external sponsorship or funding from financial entities to carry out this study-article.

Authorship contribution Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Writing (original draft and revision): Ayrtón Eduardo Wilfredo Quiñones Jara.

Availability of deposited data Not applicable.

Referencias bibliográficas Barka-Papadimitriou, A., Lasaridi, K., & Boutsouki, C. (2025). A Machine Learning Approach for the Completion, Augmentation and Interpretation of a Survey on Household Food Waste Management. *Processes*, 14(2), 302. <https://www.mdpi.com/2227-9717/14/2/302> Concha, S. M. F. (2026). Improving a Bakery's Service Level Using Machine Learning, Process Standardization, and Packaging Redesign. *Revista de la Universidad de Lima*. <https://revistas.ulima.edu.pe/index.php/CIII/article/view/8655> Feizabadi, J. (2022). Machine learning demand forecasting and supply chain performance. *International Journal of Logistics Research and Applications*, 25(2), 119-142. <https://doi.org/10.1080/13675567.2020.1803246> Feng, J., et al. (2025). Exploring the broad spectrum of machine learning technologies in the food sector: A comprehensive review of techniques and practical applications. *Food Control (PMC)*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC13091932/> Henao, J. D. V. (2023). Emerging trends in Retail analytics: a bibliometric analysis of the

Control (PMC). <https://pmc.ncbi.nlm.nih.gov/articles/PMC13091932/> Henao, J. D. V. (2023). Emerging trends in Retail analytics: a bibliometric analysis of the last decade. *Dialnet*. <https://dialnet.unirioja.es/descarga/articulo/10225362.pdf> Huber, J., & Stuckenschmidt, H. (2020). Daily retail demand forecasting using machine learning with emphasis on calendric special days. *International Journal of Forecasting*, 36(4), 1420-1438. <https://doi.org/10.1016/j.ijforecast.2020.02.005> İçöz, A. (2024). Perspective Chapter: The Role of Artificial Intelligence in Transforming the Food Industry. *IntechOpen*. <https://www.intechopen.com/online-first/1246432> Kroupová, Z. Ž. (2024). Transforming food industry and supply chain through digital innovation to reduce food waste: a systematic review. *Renewable Agriculture and Food Systems*, 39.

<https://www.cambridge.org/core/journals/renewable-agriculture-and-food-systems/article/transforming-food-industry-and-supply-chain-through-digital-innovation-to-reduce-food-waste-a-systematic-review/B03FDCB86FEEFD07E183E35C7D291F08> Mishra, H. (2024). AI-Based Predictive Waste Reduction System in Supermarkets: A Dashboard-Driven Approach. *International Journal of Science and Technology*, 3(2). https://www.ijstjournal.com/wp-content/uploads/journal/published_paper/volume-3/issue-2/IJST241239.pdf Monferdini, L. (2024). Strategies to Reduce Food Waste in the Perishable Supply Chain: A Combined Interpretive Structural Modeling and MICMAC Analysis. *RiuNet*. <https://riunet.upv.es/bitstreams/51b92486-65d2-4d51-af4f-4641de9a17fa/download> Pandey, S. (2023). Food waste management and artificial intelligence adoption in supply chain operations reference model. *RAIS Conference Proceedings*. https://scholarship.libraries.nyu.edu/view/pdfCoverPage?instCode=01RUT_INST&filePid=13770009870004646&download=true Samuels, A. (2025).

Digital transformation in supply chains: improving resilience and sustainability through AI, Blockchain, and IoT. *Frontiers in Sustainability*, 6. <https://doi.org/10.3389/frsus.2025.1584580> Wafi, M. A. A. (2025). Harnessing Artificial Intelligence (AI) to Mitigate Food Waste: Innovative Strategies for Sustainable Consumption. ResearchGate. https://www.researchgate.net/publication/387581973_Harnessing_Artificial_Intelligence_AI

Innovative Strategies for Sustainable Consumption. ResearchGate. https://www.researchgate.net/publication/387581973_Harnessing_Artificial_Intelligence_AI_to_Mitigate_Food_Waste_Innovative_Strategies_for_Sustair Wikamulia, N., Jonathan, M., & Isa, S. M. (2023). Bakery Demand Forecasting Using XGBoost and K-Means Clustering. *ICIC Express Letters, Part B: Applications*, 14(1), 21-28. <http://www.icicelb.org/ellb/contents/2023/1/elb-14-01-03.pdf> Windrasari, S. N., et al. (2023). Predictive Sales Analysis in Coffee Shops Using the Random Forest Algorithm. *Journal IRPI*. <https://journal.irpi.or.id/index.php/malcom/article/download/2023/1025/11181> Note: To strictly comply with the regulation of a minimum of 30 references, it is suggested to incorporate 15 additional complementary sources, preferably from the regional theoretical framework that accompanies your thesis at the UNT.